



Indywidualny Plan Badawczy (IPB) Individual Research Plan (IRP)

1. Podstawowe dane / Basic data

Podstawowe informacje dotyczące doktoranta i realizowanej rozprawy doktorskiej. W przypadku promotora spoza Politechniki Wrocławskiej należy podać afiliację. / *Basic information on the doctoral student and the doctoral dissertation underway. In the case of a supervisor from outside Wrocław University of Science and Technology, affiliation should be provided.*

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Dyscyplina kształcenia / Discipline of education	informatyka techniczna i telekomunikacja / information and communication technology
Wydział / Faculty	W4 Wydział Informatyki i Telekomunikacji / Faculty of Information and Communication Technology
Katedra / Department	K46 Katedra Sztucznej Inteligencji / Department of Artificial Intelligence
Data rozpoczęcia kształcenia / Date of commencement of the doctoral education	01.10.2025
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2. Temat rozprawy doktorskiej / Topic of doctoral dissertation

Propozycja tematu rozprawy doktorskiej. Przedstawiony temat nie musi stanowić ostatecznego tytułu, jaki zostanie nadany rozprawie doktorskiej złożonej na zakończenie kształcenia w Szkole Doktorskiej. / *Proposed topic for a doctoral dissertation. The presented topic does not have to constitute the final title of the doctoral dissertation submitted at the end of education at the Doctoral School.*

Improving the generalization of deep learning models in real-to-sim-to-real transfer for physical AI.
(PL: Poprawa zdolności generalizacji modeli uczenia głębokiego w transferze rzeczywistość–symulacja–rzeczywistość dla fizycznej sztucznej inteligencji.)

3. Harmonogram przygotowania rozprawy doktorskiej / Schedule for the preparation of the doctoral dissertation

Harmonogram przygotowania rozprawy doktorskiej z wyszczególnieniem etapów i miejsc prowadzonych badań, w tym terminów zakończenia badań cząstkowych (semestr kształcenia) i opracowania ich wyników oraz opcjonalnie w zależności od rodzaju prowadzonych badań, wybranych między innymi spośród następujących:

- realizacja badań (w tym przygotowanie stanowiska badawczego, przygotowanie materiałów do badań, wykonanie pomiarów/eksperymentów, opracowanie otrzymanych wyników),
- przygotowanie i wysłanie artykułu naukowego do redakcji czasopisma naukowego,



- wysłanie zgłoszenia patentowego do właściwego urzędu,
- opracowanie propozycji wdrożenia dla przemysłu,
- udział w konferencjach, warsztatach lub szkołach letnich/zimowych i prezentacja wyników badań,
- przygotowanie wniosku i wystąpienie o grant,
- mobilność krajowa lub zagraniczna doktoranta (w tym konsultacje naukowe i wizyty studyjne w ośrodkach krajowych lub zagranicznych).

A schedule for the preparation of the doctoral dissertation detailing the stages and locations of the research to be conducted, including the dates for completing the sub-studies and developing their results, and optionally, depending on the type of research to be conducted, selected among the following:

- *execution of the research (including preparation of the research site, preparation of materials for the research, execution of measurements/experiments, elaboration of the obtained results),*
- *preparation and submission of a scientific article to the editor of a scientific journal,*
- *submission of a patent application to the relevant office,*
- *development of an implementation proposal for the industry,*
- *participation in conferences, workshops or summer/winter schools and presentation of research results,*
- *preparation of an application and application for a scholarship,*
- *domestic or foreign mobility of the doctoral student (including scientific consultations and study visits to domestic or foreign centres).*

Okres realizacji zadania (semestr kształcenia) / Task implementation period (semester)	Zwięzły opis zadania / Brief description of the task
Semestr 1	1. Literature review in the scope of sim-to-real transfer, domain randomization and domain adaptation, neural scene reconstruction from real data (such as 3D Gaussian Splatting) and deep learning for robot navigation and manipulation. 2. Identification of the most important research gaps in the generalization of deep learning models trained in simulations built from real data. 3. Extending theoretical knowledge, including machine learning theory and learning under distribution shift, necessary for further work on the PhD thesis. 4. Participation in the Doctoral School courses.
Semestr 2	1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI. 2. Initial pilot experiments on reconstructing real scenes with 3D Gaussian Splatting and importing them into a navigation simulator, with regard to the gaps identified during the literature review. 3. Specifying the Individual Research Plan. 4. Participation in the Doctoral School courses.



Semestr 3	1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI. 2. Preparation of the research environment, including real scene captures, simulations built from them, a manipulation environment with hidden visual and physical parameters, and computing resources. 3. Conducting research on which components of the discrepancy between a simulation built from real data and reality determine the generalization of models trained in it. 4. Carrying out initial analysis of where inside a model trained in simulation the simulation-to-reality gap arises.
Semestr 4	1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI. 2. Continuing and finishing the research on the components of the simulation-to-reality discrepancy, including physical components in the manipulation setting, and on shaping the training distribution to target the harmful components. 3. Writing a scientific article for a conference from the ministerial list (regarding the components of the simulation-to-reality discrepancy that determine generalization). 4. Continuing the research on localizing the simulation-to-reality gap inside the model and on task-relevant invariant representations. 5. Preparation of the Preludium research grant proposal to the National Science Centre. 6. Participation in an international scientific conference.
Semestr 5	1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI. 2. Finishing the research on localizing the simulation-to-reality gap and on task-relevant invariance at that location. 3. Writing a scientific article for a conference from the ministerial list (regarding the localization of the simulation-to-reality gap inside deep models). 4. Carrying out research on selecting real data under a limited budget to correct both the simulation and the model. 5. Initial validation of the developed navigation methods on a real mobile robot.
Semestr 6	1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI. 2. Continuing and finishing the research on the selection of real data for correcting both the simulation and the model. 3. Writing a scientific article for a conference from the ministerial list (regarding the selection of real data under a limited budget). 4. Research on the generalization of the developed methods to unseen environments and to robotic manipulation as a controlled second task. 5. Research internship at a foreign research group working on robot learning and sim-to-real transfer.



Semestr 7	1. Continuation of the literature review - participation in the internal reading group, discussing novel methods regarding sim-to-real transfer and physical AI. 2. Finishing the studies on the generalization of the developed methods across environments and tasks, including final validation of navigation on a real robot. 3. Writing a scientific article for a conference from the ministerial list (regarding which properties of the shift explain when improvements generalize). 4. Participation in an international scientific conference. 5. Preparing the structure and sketch of the doctoral dissertation.
Semestr 8	1. Preparation of a journal article from the ministerial list summarizing the results on the generalization of the developed methods across environments and tasks. 2. Editing the doctoral dissertation, preparation of the final version of the document.

4. Termin złożenia rozprawy doktorskiej / Date of the doctoral dissertation submission

Planowany termin złożenia rozprawy doktorskiej nie może przekraczać daty zakończenia ósmego semestru kształcenia w Szkole Doktorskiej. / A planned date for submission of the dissertation, which must not exceed the date of completion of the eighth semester of education in the Doctoral School.

September 2029

(rekomendowana data to ostatni dzień z okresu 4 lata kształcenia w Szkole Doktorskiej, np. jeżeli kształcenie rozpoczęło się 1.10.2025, to proszę wpisać 30.09.2029 /

The recommended date, which marks the final day of the four-year doctoral training period. For example, if your training began on October 1st, 2025, please enter September 30th, 2029)

5. Uzasadnienie wyboru tematu rozprawy doktorskiej / Justification for choosing the topic of the doctoral dissertation

Opis zawierający uzasadnienie wyboru tematu rozprawy doktorskiej uwzględniający potencjalne obszary zastosowania planowanych wyników rozprawy doktorskiej (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description containing the justification for the selection of the topic of the doctoral dissertation, taking into account the potential areas of application of the planned results of the doctoral dissertation (at most 1 page, font 11, line spacing 1).

Deep learning has brought remarkable progress in computer vision and natural language processing, largely thanks to vast amounts of data collected from the web. Physical AI, i.e., AI in robots and other embodied systems that perceive and act in the real world, cannot rely on such data. Its models need interaction data, which are slow, expensive and sometimes unsafe to collect with real robots. Therefore, such models are commonly trained in simulation. Recently, simulations built from real data have become a promising alternative to hand-crafted simulators: neural scene reconstruction methods, such as 3D Gaussian Splatting, turn a short recording of a real place into a photorealistic digital twin in which a model can be trained at scale. This forms a real-to-sim-to-real loop, in which real data are turned into a simulation, a model is trained there, and the model is transferred back to reality.



However, the value of this loop depends on how well the trained models generalize. A model that performs well in simulation often fails in reality, since no simulation reproduces the real world exactly: appearance, geometry, lighting and physics are only approximated, and reconstruction errors differ between regions of a scene. This simulation-to-reality generalization gap is caused by the shift between the distributions of simulated and real data, which makes it a particular case of learning under distribution shift and a central obstacle to physical AI. The methods proposed so far have several limitations. Domain randomization, even when fitted to real data, varies a few global parameters regardless of where the simulation is inaccurate, and invariant representations may discard information needed for the task. Moreover, the real data used for adaptation are usually chosen at random, by hand or by simple heuristics such as observed failures, and serve to correct either the model or the simulation, but not both. Finally, most methods are evaluated on a single task and a few scenes, and it is not known which properties of the simulation and of the learned representations determine whether a model generalizes to reality.

Therefore, the dissertation aims to improve the generalization of deep learning models in real-to-sim-to-real transfer at the points of the loop where the gap arises. The research shall focus on understanding which errors of a simulation built from real data are harmful to generalization, on localizing where inside a model the gap arises and reducing it there, on choosing the small amount of real data that corrects both the simulation and the model most efficiently, and on verifying whether the obtained improvements carry over to unseen environments and to another physical task, and which properties of the shift explain when they do. Robot navigation shall serve as the main testbed and robotic manipulation as the second task. Beyond more reliable models, the research could help explain why models trained on synthetic data succeed or fail on real data, which is relevant to machine learning in general.

The results have potential applications in service, logistics and assistive robotics, where a robot could be trained in a digital copy of a new warehouse, hospital or home and operate there reliably after little real-world adaptation. More broadly, they concern any domain in which models are trained on simulated or reconstructed data and deployed on real sensor data, such as autonomous driving or inspection robots.

6. Zarys aktualnego stanu badań w tematyce rozprawy doktorskiej / **Outline of the current state of research in the topic of the doctoral dissertation**

Przegląd literaturowy zawierający opis aktualnego stanu badań w tematyce rozprawy doktorskiej uwzględniający odwołania do najważniejszych pozycji literaturowych związanych z tematyką rozprawy doktorskiej (maks. 2 strony, czcionka 11 pt., interlinia 1). / A literature review containing a description of the current state of research in the subject of the doctoral dissertation, including references to the most important literature items related to the subject of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

Deep learning models for navigation and manipulation need more experience than real robots can provide, so they are usually trained in simulation. However, performance in simulation can be an unreliable predictor of real-world performance unless the simulator is carefully tuned [1], and higher fidelity alone does not ensure better transfer [2]. This is an instance of learning under distribution shift: domain adaptation theory bounds the error of a model on the target domain by its error on the source domain and the discrepancy between the two distributions [3]. This suggests two levers: a training distribution that covers reality, and a model insensitive to the remaining discrepancy.

The first lever is exploited by domain randomization, which varies simulator parameters such as textures so that reality appears as one more variation [4]. Since overly wide randomization leads to conservative behaviour, later works shape its distribution: they fit it to a few real rollouts [5], infer a posterior over simulator parameters [6], or maximize its entropy while preserving task success [7]. Yet they act on a few global parameters of a synthetic simulator, not on errors that vary across a scene.

The second lever is invariance. Domain-adversarial training aligns features across domains [8]. In robotics, randomized and real images were mapped to a canonical image [9], and aligning the joint distributions of observations and actions in simulated and real data improved co-trained policies in reality [10]. However, invariance with low source error does not guarantee transfer [11], and full invariance may discard task information. Moreover, it is usually enforced at a fixed location, the input or final features.

A recent development is building simulations from real data. 3D Gaussian Splatting [12] reconstructs photorealistic scenes from ordinary images, and digital twins built from real data have been used to train manipulation policies from a scan of the target scene [13] and navigation policies from a phone capture of a room [14]. In twins, uncertainty has so far been modelled mainly for physical parameters [15]; beyond this, a twin is typically treated as a fixed copy of reality. It is unknown which of its errors, in appearance, geometry, lighting or physics, harm generalization, and existing training variation ignores how uncertain the reconstruction is in each region.

It is also unclear where inside a model the gap arises. Self-supervised features are robust to many appearance changes [16], yet no representation dominates across navigation, manipulation and locomotion [17], and manipulation encoders degrade under lighting and texture shifts [18]. Probing showed that action fine-tuning degrades the visual representations of robot policies [19], and the best layers to adapt depend on the type of shift [20], but neither localizes the gap in models trained in reconstructed simulations.

The remaining gap is usually reduced with little real data. Co-training with a small real set yields large gains [21], and active system identification collects the real trajectories most informative about physics [22]. In vision, selecting images to label by domain discrepancy and uncertainty beats random selection [23]. The closest work uses a twin to find failure-prone configurations for real rollouts [24]. However, robotic methods correct either the model or the simulation and choose real data at random, by hand, at the model's failures, or by their informativeness about physical parameters; to the best of our knowledge, none uses the reconstruction uncertainty of a simulation built from real data, or the same data to correct both.

Finally, generalization to unseen scenes depends on training-scene diversity [25], and the gap can be decomposed into factors of variation whose difficulty ordering is largely consistent between simulation and reality [26]. However, it is rarely studied whether a sim-to-real method keeps its benefit when the scene or the task changes.

In summary, it is open which simulation errors govern generalization, where inside a model the gap arises, which real data best correct model and simulation, and when improvements hold in new environments and tasks; these gaps motivate the questions in Section 7.

References

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- [25] Deitke, M., et al. (2022). ProcTHOR: Large-scale embodied AI using procedural generation. NeurIPS.
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7. Pytania i hipotezy badawcze / Research questions and hypotheses

Opis najważniejszych pytań i hipotez badawczych związanych z realizowaną rozprawą doktorską (maks. 1 strona, czcionka 11 pt., interlinia 1). / Description of the most important research questions and hypotheses related to the doctoral dissertation (at most 1 page, font 11, line spacing 1).

The research carried out in the proposed doctoral dissertation is aimed at studying the following research questions.

1. **Which components of the discrepancy between a simulation built from real data and reality determine how well models trained in it generalize?** Domain adaptation theory bounds the real-world error of a model by its error in simulation and by the discrepancy between the simulated and the real distributions. A simulation built from real data reproduces appearance, geometry, lighting and physics only approximately, and its errors differ between regions of a scene. It is hypothesized that the harm caused by a simulation error is predicted better by the change it induces in the task-relevant representations of the model than by its magnitude in the image or geometry, so that errors of similar size differ strongly in their effect on generalization. Hence, training variation should follow the reconstruction uncertainty and task relevance of each region rather than be spread uniformly. The research will attribute the gap to types of simulation error and derive a principled way to shape the training distribution accordingly.
2. **Where inside a model trained in simulation does the simulation-to-reality gap arise, and can task-relevant invariance at that location reduce it?** Models trained in simulation can rely on features absent in reality, such as reconstruction artifacts. Invariant representations reduce the discrepancy between domains, yet full invariance does not guarantee transfer and may remove



information needed for the task. Moreover, it is not known at which stage of such a model, from early visual features to the action output, the shift turns into errors. It is hypothesized that the gap arises in a limited part of the model, and that enforcing invariance there, only for task-relevant features and from a few paired simulated and real observations, yields better generalization than invariance at the input or at the final features. The research will use probing of hidden states to localize the gap before intervening.

3. **Which real data, chosen under a limited budget, most efficiently correct both the simulation and the model, and thereby reduce the generalization gap?** Real interaction data, such as robot trials or demonstrations, are expensive to collect. Existing robotic methods correct either the model or the simulation and choose real data at random, by hand, where the model fails, or by their informativeness about physical parameters; selection by domain discrepancy has so far been studied only for labelling images. It is hypothesized that the most informative real data are those where the simulation is predicted to disagree with reality in task-relevant ways, as estimated from its reconstruction uncertainty, the distance between the task-relevant representations of simulated and cheaply captured real observations, and the sensitivity of the predicted action. Using the selected data to correct both the simulation and the model is expected to reduce the gap with less real data than random or failure-driven selection. The research will develop such a selection rule and study it under different budgets.
4. **Do the improvements generalize to unseen environments and to another physical task, and which properties of the shift explain when they do?** A valuable method should not be tied to a single scene or task, and no single visual representation is known to be best for both navigation and manipulation. Therefore, the developed methods shall be examined, with unchanged settings, on environments held out from training and on two physical tasks: robot navigation and, as a controlled second task, robotic manipulation. It is hypothesized that the attribution of the gap from the first question and its localization from the second, measured in the new setting before a method is applied, predict whether its improvement transfers. The research will decompose the gap into factors of variation and relate the measured attribution and localization to the observed transfer across held-out scenes and tasks.

8. Wkład spodziewanych wyników dla rozwoju dyscypliny naukowej / Contribution of the expected results to development of scientific discipline

Krótki opis spodziewanego wkładu wyników rozprawy doktorskiej dla rozwoju dyscypliny naukowej, w której realizowana jest rozprawa doktorska (maks. 1 strona, czcionka 11 pt., interlinia 1). / Brief description of the expected contribution of the results of the doctoral dissertation to the development of the scientific discipline in which the doctoral dissertation is carried out (at most 1 page, font 11, line spacing 1).

The proposed PhD dissertation aims to introduce several contributions to the development of the scientific discipline. The central scientific problem addressed is explaining and reducing the generalization gap caused by the distribution shift between a simulation built from real data and reality; poor generalization of models trained in simulation remains one of the main obstacles to applying machine learning in physical AI. The central claim of the dissertation is that this gap is concentrated in a small set of task-relevant discrepancies, which can be identified in the simulation, localized inside the model and targeted with little real data, and that addressing them there yields better generalization than uniform randomization, invariance of the whole representation or randomly chosen real data. Research questions 1-3 test the three parts of this claim, and question 4 tests whether it holds across environments and tasks; the methods derived from it are intended to be general, i.e., not tied to a particular robot, scene or task, and will be examined on robot navigation and robotic manipulation.

Furthermore, the research is expected to deliver new methods and findings that do not exist at present: a principle for shaping the training distribution of a simulation built from real data by the reconstruction uncertainty and task relevance of each region; a method for localizing the simulation-



to-reality gap inside a deep model and for enforcing task-relevant invariance only there; a rule for selecting real data that corrects the simulation and the model at the same time; and an empirical account of which properties of the shift determine whether these improvements carry over across environments and tasks. Such knowledge could complement the currently dominant randomization heuristics with more principled and interpretable approaches, and could introduce new perspectives for the design of simulations and of training procedures.

Since transfer from simulation to reality is an instance of learning under distribution shift, the findings are also expected to be relevant to machine learning in general. The analysis of which errors in the training distribution are harmful concerns domain adaptation; the localization of the simulation-to-reality gap inside a model extends probing methods developed for language and vision models and contributes to representation learning; and the selection of informative data under a budget concerns active learning and the adaptation of pretrained foundation models to new domains from a few examples.

Finally, the developed methods could reduce the amount of real data, and thus the cost, needed to deploy learning-based robots in new environments. The source code and the experimental setups are planned to be released, which could facilitate further research in the field and attract other researchers. Apart from purely scientific results, the methods could be used in real-world applications such as service, logistics and inspection robotics, as well as in other domains where models are trained in simulation, e.g., autonomous driving.

9. Planowane metody badawcze / Planned research methods

Krótki opis metod badawczych, które są będą użyte w czasie realizacji rozprawy doktorskiej dla (maks. 2 strony, czcionka 11 pt., interlinia 1). / Brief description of the research methods that will be used during the implementation of the doctoral dissertation (at most 2 pages, font 11, line spacing 1).

The research will follow the methods commonly adopted in machine learning and robot learning projects. Since no established theory guarantees generalization from simulation to reality, the research will be mainly empirical. The proposed methods will be described theoretically, and their properties or bounds will be analysed whenever possible, in particular using domain adaptation theory, which relates the error of a model in reality to its error in simulation and to the discrepancy between the two distributions. Numbers in brackets refer to the references listed in Section 6.

Simulations will be built from real captures of indoor scenes, using structure-from-motion for camera poses and neural scene reconstruction, such as 3D Gaussian Splatting, for appearance and geometry, and then coupled with robot simulators, e.g., Habitat for navigation and ManiSkill3 for manipulation, with geometry for collisions and physics and appearance rendered from the reconstruction or from a mesh extracted from it. Reconstruction uncertainty will be estimated per region, e.g., from view coverage and the disagreement of an ensemble of reconstructions. Models for goal-directed indoor navigation (e.g., point-goal or image-goal) and tabletop pick-and-place manipulation will be trained with reinforcement and imitation learning, starting from pretrained models when possible.

Experiments will be conducted mainly on public datasets of real scenes, such as ScanNet++ and MuSHRoom, which contain separate captures of the same scenes made with different devices. This allows building the simulation from one capture and using another, independent capture as a proxy for reality. Real frames of the independent capture will serve for open-loop evaluation, which measures the discrepancy between representations of paired simulated and real views, the agreement of predicted actions, and probing accuracy. A reference simulation built from the independent, higher-fidelity capture (e.g., the laser scan and DSLR images in ScanNet++) will serve as a proxy for closed-loop evaluation. Thus, the gap can be studied on many scenes without a robot. For manipulation, where real data with known physical properties are scarce, a simulation whose visual and physical parameters are hidden from the learner will serve as a controlled proxy.



For the first question, controlled variants of a simulation, in which components such as appearance, geometry or lighting are degraded or replaced, will be used to attribute the gap to types of simulation error; physical components will be studied in manipulation, where they can be controlled. The task relevance of a region will be estimated from the sensitivity of the model's actions to it, e.g., with gradient-based attribution or by occluding the region. The training distribution will then be shaped by randomizing appearance (colour and opacity of Gaussians), geometry and, where the representation allows relighting, lighting per region in proportion to its reconstruction uncertainty and task relevance, and compared with uniform randomization, randomization fitted to real rollouts [5] and entropy-maximizing randomization [7]. For the second question, probing of hidden states will be adapted to find at which layers simulated and real observations become distinguishable and where task-relevant information is lost. An alignment loss (e.g., distribution-discrepancy or contrastive) will then be applied at the identified layers to a few paired simulated and real views at the same camera poses, restricted to the features on which the predicted action depends. It will be compared with alignment at the input, domain-adversarial alignment of the final features [8] and full fine-tuning.

For the third question, the budget refers to real interaction data, such as robot trials or demonstrations, while unlabelled real images are assumed available. On the dataset-based proxy, a unit of interaction data will be an expert or policy episode executed in the reference simulation, paired with the real frames of the independent capture at the visited poses; in manipulation, an episode in the simulation with hidden parameters; and on the physical robot, a real trial or demonstration. Episodes (start and goal configurations), poses and scene regions used for final evaluation will be disjoint from those available for selection, and results will also be reported on held-out scenes whose independent capture is never used for correction. Real data will be selected by scores combining reconstruction uncertainty, the distance between the task-relevant representations of simulated and real observations, and the sensitivity of the predicted action. Each selected location will thus yield both real views, used to refine the simulation by re-optimizing the reconstruction in uncertain regions (and, in manipulation, by re-estimating physical parameters), and interaction data, used to adapt the model by fine-tuning or co-training on simulated and real data. This will be compared with random and failure-driven selection, in manipulation also with selection by informativeness about physics [22], with the same score without task relevance, and with correcting only the model or only the simulation, at equal amounts of real data across budgets. For the fourth question, the methods will be applied with unchanged settings to held-out scenes and to manipulation, the gap will be decomposed into factors of variation, and the attribution and localization measured in a new setting will be related, across many held-out scenes and both tasks, to the size of the improvement there, e.g., by rank correlation.

The quantitative evaluation will rely on standard metrics for embodied tasks, such as success rate and success weighted by path length (SPL), and on the generalization gap, defined at two levels: on the dataset-based proxy, as the difference between the performance in the training simulation and in the reference simulation, complemented by the open-loop measures on real frames; and on the physical robot, as the difference between the closed-loop performance in simulation and in reality. The agreement between the two levels will also be measured, e.g., by the rank correlation of model scores. All methods will be compared with baselines under equal training effort and an equal amount of real data. Statistical methods will be leveraged to provide reliable outcomes, including repeated experiments with different random seeds and scenes, confidence intervals, and statistical tests for comparisons with baseline methods. According to accepted standards, the data used for training, model selection and final evaluation will be kept separate. The analysis of the learned representations will combine quantitative and qualitative methods: representations of paired simulated and real observations will be compared with representation similarity measures, visualized in the latent space, and related to failure cases.

The navigation results will be validated on a wheeled mobile robot with an RGB-D camera, e.g., in the robotics laboratories of the university. The validation will measure closed-loop success rate and SPL in reality and check whether the conclusions drawn with the dataset-based proxy also hold on a physical robot. Should robot access be limited, the conclusions will rest on the dataset-based proxy and on a



reduced set of real trials, and the agreement between the two levels will be reported for the trials performed.

Experiments will be implemented mainly in Python with the PyTorch deep learning library. Source code will be managed with the Git version control system, and datasets, configurations and results will be versioned and tracked with experiment tracking tools to provide reproducible results. Due to the high computational demand of scene reconstruction and policy training, the experiments will run on GPU clusters provided by the Department and on the Polish computing infrastructure (WCSS and PLGrid). The work will be partly conducted in cooperation with other PhD students and senior researchers to increase the quality of the research. Code and data will be made publicly available whenever possible, following open science standards and respecting the licences of the datasets used.

The findings will be published at leading international machine learning, computer vision and robotics conferences, such as NeurIPS, ICML, ICLR, CVPR and RSS, and in journals from the ministerial list. To promote the work and establish new collaborations, it is planned to present results at smaller conferences and workshops, such as MLinPL, and at summer schools, and to seek foreign co-authorship during the research internship.

10. Streszczenie popularnonaukowe / Abstract for general public

Streszczenie popularnonaukowe należy przygotować w dwóch językach: polskim i angielskim. Wersje językowe muszą być tożsame. Opis powinien być napisany językiem popularnonaukowym i zawierać cel rozprawy, opis planowanych badań naukowych oraz powodów, dla których podjęta została ta tematyka badawcza, a także najważniejsze spodziewane efekty (maks. 1 strona, czcionka 11 pt., interlinia 1). / The description for general public must be in Polish and in English. The language versions must be identical. The description should be written for general public and should include the doctoral dissertation goal, description of research, reasons for attempting a particular research topic and substantial results expected (at most 1 page, font 11, line spacing 1).

Streszczenie popularnonaukowe

Sztuczna inteligencja coraz częściej wychodzi poza ekrany komputerów i trafia do systemów fizycznych, takich jak roboty poruszające się po budynkach lub przenoszące przedmioty. Modele uczenia głębokiego sterujące takimi robotami potrzebują ogromnej liczby przykładów, których zebranie w prawdziwym świecie jest powolne, kosztowne, a czasem niebezpieczne. Dlatego modele te uczą się zwykle w symulacji, a coraz częściej w cyfrowych bliźniakach, czyli wirtualnych kopiach rzeczywistych miejsc odtworzonych na podstawie ich nagrań. Taka kopia nigdy nie jest jednak idealna, a model, który dobrze działa w symulacji, w rzeczywistości często zawodzi, zwłaszcza w miejscach, których wcześniej nie widział, co pozostaje jedną z głównych przeszkód na drodze do szerszego zastosowania fizycznej sztucznej inteligencji. Celem rozprawy jest poprawa zdolności generalizacji modeli uczenia głębokiego trenowanych w symulacjach zbudowanych na podstawie danych rzeczywistych, czyli umiejętności poprawnego działania w warunkach innych niż te, w których je trenowano, a zwłaszcza w rzeczywistości. Badania w pierwszej kolejności pozwolą ustalić, które różnice między symulacją a rzeczywistością faktycznie szkodzą modelowi i na którym etapie przetwarzania w modelu przekształcają się one w błędy. Na tej podstawie zostaną opracowane metody, które nauczą model pomijać nieistotne różnice i więcej ćwiczyć tam, gdzie wirtualna kopia jest najmniej dokładna, a także metody wyboru nielicznych rzeczywistych przykładów, które najlepiej poprawiają zarówno model, jak i symulację. Metody zostaną sprawdzone w zadaniach nawigacji robotów i manipulacji robotycznej, w środowiskach, które nie były wykorzystywane podczas uczenia, a nawigacja zostanie zweryfikowana na prawdziwym robocie mobilnym. Oczekuje się, że rozprawa przyniesie lepsze zrozumienie przyczyn, dla których modele uczone w symulacji zawodzą w rzeczywistości, a także metody, dzięki którym roboty będą mogły szybciej, taniej i bezpieczniej uczyć się pracy w nowych miejscach i przy różnych zadaniach.



Abstract for general public

Artificial intelligence is increasingly moving beyond computer screens into physical systems, such as robots that move around buildings or carry objects. Deep learning models that control such robots need a huge number of examples, which are slow, costly and sometimes unsafe to collect in the real world. Therefore, such models are usually trained in simulation, and increasingly in digital twins, that is, virtual copies of real places rebuilt from recordings of those places. Such a copy is never perfect, however, and a model that works well in simulation often fails in reality, especially in places it has not seen before, which remains one of the main obstacles to the wider use of physical artificial intelligence. The goal of the dissertation is to improve the generalization of deep learning models trained in simulations built from real data, i.e., their ability to work correctly in conditions different from those they were trained in, and in particular in reality. The research will first establish which differences between simulation and reality actually harm the model, and at which stage of processing inside the model they turn into errors. On this basis, methods will be developed that teach the model to ignore irrelevant differences and to practise more where the virtual copy is least accurate, together with methods that choose the few real examples that best correct both the model and the simulation. The methods will be examined on robot navigation and robotic manipulation tasks, in environments that were not used during training, and navigation will be verified on a real mobile robot. The dissertation is expected to bring a better understanding of why models trained in simulation fail in reality, as well as methods that allow robots to learn to work in new places and on different tasks faster, more cheaply and more safely.

11. Termin oddania do druku artykułu naukowego lub monografii / Date of submission for publication of at least one scientific article or monography

Termin wysłania do recenzji artykułu naukowego lub monografii naukowej zgodnie z wymaganiami art. 186 ust. 1 pkt 3 Ustawy stawianymi osobie ubiegającej się stopień doktora. Termin dotyczy oddania do druku co najmniej jednego artykułu naukowego opublikowanego w czasopiśmie naukowym lub w recenzowanych materiałach z konferencji międzynarodowej w odpowiedniej dyscyplinie naukowej, które w roku opublikowania artykułu w ostatecznej formie były ujęte w wykazie („lista ministerialna”) sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. b) lub 1 monografii naukowej wydanej przez wydawnictwo, które w roku opublikowania monografii w ostatecznej formie było ujęte w wykazie sporządzonym zgodnie z przepisami wydanymi na podstawie Ustawy (art. 267 ust. 2 pkt 2 lit. a). / A date of submission for publication of at least one scientific article published in a scientific journal or in the peer-reviewed materials of an international conference in the relevant scientific discipline, which in the year of publication of the article in its final form were included in the list drawn up in accordance with the regulations issued under the Act (Art. 267 sec. 2 point 2(b) or 1 scientific monograph published by a publishing house, which in the year of publication of the monograph in its final form was included in the list drawn up in accordance with the regulations issued pursuant to the Act (Art. 267 sec. 2 point 2(a), referred to in Art. 186 sec. 1 point 3 of the Act.

May 2027

12. Inne / Other comments

Dodatkowe wyjaśnienia, komentarze, podsumowanie związane z indywidualnym planem badawczym (maks. 1 strona, czcionka 11 pt., interlinia 1). / Additional explanations, comments, summary related to the individual research plan (at most 1 page, font 11, line spacing 1).

13. Określenie planowanej formy współpracy z promotorem / Planned form of cooperation with the supervisor



Krótki opis planowanej formy współpracy z promotorem, np. częstotliwość spotkań, forma spotkań i ich zakres, sposób komunikacji. / Brief description of the planned form of cooperation with the supervisor, e.g., frequency of meetings, form of meetings and their scope, method of communication.

It is planned to have weekly meetings with the supervisor. The meetings are planned to consist of results presentations, discussions, reviews and joint work on papers. The scope of meetings is also supposed to include the evaluation of progress and the planning of further activities. Moreover, it is planned to discuss results, issues and recent advancements in the field with other Ph.D. students weekly, with the supervisor's participation. Day-to-day communication will take place by e-mail and instant messaging.

14. Opinia promotora pomocniczego / Opinion of assistant supervisor

Promotor pomocniczy (jeżeli jest wyznaczony) powinien zaopiniować IPB. / Assistant supervisor (if appointed) should provide an opinion of the IRP.

nie dotyczy / not applicable

.....
Podpis promotora pomocniczego /
Signature of assistant supervisor

15. Podpisy doktoranta i promotora/promotorów / Signatures of doctoral student and supervisor(s)

Wybierz datę/Select date

.....
Data / Date

.....
Podpis doktoranta / Signature of doctoral student

.....
Podpis promotora / Signature of supervisor

.....
Podpis drugiego promotora / Signature of second supervisor